```
import sqlite3
import pandas as pd
from IPython.display import display

# 1. Download real E-commerce data (Northwind Dataset) directly from GitHub
print("Downloading real dataset...")
url_customers = 'https://raw.githubusercontent.com/graphql-compose/graphql-compose-examples/master/examples/northwind/data/c
url_orders = 'https://raw.githubusercontent.com/graphql-compose/graphql-compose-examples/master/examples/northwind/data/csv/
```

```
Downloading real dataset...
```

```
customers_df = pd.read_csv(url_customers)
orders_df = pd.read_csv(url_orders)
```

```
conn = sqlite3.connect('ecommerce_real.db')
customers_df.to_sql('Customers', conn, if_exists='replace', index=False)
orders_df.to_sql('Orders', conn, if_exists='replace', index=False)
print("Database created successfully!\n")
```

```
Database created successfully!
```

```
query_a = "SELECT companyName, city, country FROM Customers WHERE country = 'USA' ORDER BY companyName LIMIT 5;"
display(pd.read_sql_query(query_a, conn))
```

	companyName	city	country
0	Great Lakes Food Market	Eugene	USA
1	Hungry Coyote Import Store	Elgin	USA
2	Lazy K Kountry Store	Walla Walla	USA
3	Let's Stop N Shop	San Francisco	USA
4	Lonesome Pine Restaurant	Portland	USA

```
query_d = "SELECT customerID, SUM(freight) AS total_freight, AVG(freight) AS avg_freight FROM Orders GROUP BY customerID L:
display(pd.read_sql_query(query_d, conn))
```

	customerID	total_freight	avg_freight
0	ALFKI	225.58	37.596667
1	ANATR	97.42	24.355000
2	ANTON	268.52	38.360000
3	AROUT	471.95	36.303846
4	BERGS	1559.52	86.640000

```
query_b1 = '''
SELECT c.companyName, o.orderDate, o.freight
FROM Customers c
INNER JOIN Orders o ON c.customerID = o.customerID
LIMIT 5;
'''
display(pd.read_sql_query(query_b1, conn))
```

	companyName	orderDate	freight
0	Alfreds Futterkiste	1997-08-25 00:00:00.000	29.46
1	Alfreds Futterkiste	1997-10-03 00:00:00.000	61.02
2	Alfreds Futterkiste	1997-10-13 00:00:00.000	23.94
3	Alfreds Futterkiste	1998-01-15 00:00:00.000	69.53
4	Alfreds Futterkiste	1998-03-16 00:00:00.000	40.42

```
query_b2 = '''
SELECT c.companyName, o.orderID
FROM Customers c
LEFT JOIN Orders o ON c.customerID = o.customerID
LIMIT 5;
'''
display(pd.read_sql_query(query_b2, conn))
```

	companyName	orderID
0	Alfreds Futterkiste	10643
1	Alfreds Futterkiste	10692
2	Alfreds Futterkiste	10702
3	Alfreds Futterkiste	10835
4	Alfreds Futterkiste	10952

```
query_c = '''
SELECT companyName FROM Customers
WHERE customerID IN (SELECT customerID FROM Orders WHERE freight > 500);
'''
display(pd.read_sql_query(query_c, conn))
```

	companyName
0	Ernst Handel
1	Great Lakes Food Market
2	Hungry Owl All-Night Grocers
3	Queen Cozinha
4	QUICK-Stop
5	Rattlesnake Canyon Grocery
6	Save-a-lot Markets
7	White Clover Markets

```
cursor.executescript('''
    -- Create a View
    CREATE VIEW IF NOT EXISTS HighFreightCustomers AS
    SELECT c.companyName, SUM(o.freight) as total_freight
    FROM Customers c JOIN Orders o ON c.customerID = o.customerID
    GROUP BY c.customerID HAVING SUM(o.freight) > 1000;

    -- Create an Index for optimization
    CREATE INDEX IF NOT EXISTS idx_customer_id ON Orders(customerID);
''')
print("\n--- 6. View 'HighFreightCustomers' and Index 'idx_customer_id' created successfully! ---")

conn.commit()
```

```
--- 6. View 'HighFreightCustomers' and Index 'idx_customer_id' created successfully! ---
```